

Aryan Chahardovalee

Electrical Engineer | Flight Test Operations | RF, Embedded, FPGA, Space, Maritime & Drone Systems

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| HAM Callsign: VA7AYO

SUMMARY

Electrical engineer with 5+ years of hands-on experience across autonomous aircraft flight testing, UAS system validation, RF communications, embedded hardware, and test documentation. At Zipline, led flight-test validation within a small team supporting autonomous aircraft systems, including test execution, procedural discipline, regression analysis, troubleshooting, and operational readiness.

SKILLS

- Flight Test & Operations:** test planning, test readiness, briefings/debriefings, procedural execution, UAS preflight/recovery, SOPs, post-test reporting
- Aircraft & UAS Systems:** autonomous aircraft operations, RF communication links, telemetry awareness, aircraft readiness checks, environmental limitations, wind sensitivity, signal integrity, camera/IMU systems, GPS/Wi-Fi, drone integration
- Data & Analysis:** regression analysis, Python analysis, AI test automation, telemetry review, anomaly investigation
- Hardware & Instrumentation:** flight-test support equipment, embedded systems, RF systems, sensor integration, PCBs
- Software & Tools:** Python, C/C++, MATLAB, Linux, shell scripting, Verilog, Streamlit, Git/GitHub, data visualization

EXPERIENCE

Flight Test Operations Engineer - [Zipline International](#) 2024 to Present

- Led flight-test validation within a small team, ensuring safety, procedural discipline, and operational readiness for autonomous aircraft systems.
- Documented and analyzed test outcomes, identified regressions, communicated results to stakeholders, and supported troubleshooting through resolution.
- Developed troubleshooting guides and test procedures with multidisciplinary teams to reduce operational risk and improve test repeatability.
- Identified system vulnerabilities and operating limitations, including wind sensitivity, signal integrity issues, communication reliability, and environmental constraints.
- Added a data-system feature for marking flights as representative or non-representative, including reasons for exclusion, to reduce false negatives in regression analysis.
- Identify and document ambiguous test behavior, transforming data into measured engineering decisions.

AI/Hardware Integration Engineer - [Huawei Canada](#) 2023 to 2024

- Developed and tested drone WLAN architecture, improving system response speed by 250%.
- Implemented real-time processing for camera and LiDAR data to support obstacle detection.
- Optimized power systems, significantly extending flight duration.

Electrical Engineering Intern - [Seaspan](#) 2022

- Drafted design documents for communications systems in maritime contexts.
- Contributed to R&D efforts using advanced EMC simulation tools, achieving a 300% improvement in study efficiency.
- Assessed Wi-Fi system upgrades within extensive fleet usage scenarios.

RF & Hardware Engineer - [UBC ORBIT](#) 2021 to 2023

- Enhanced CubeSat radio systems, achieving a 40% reduction in bit error rate.
- Designed and validated PCB systems for satellite communication applications.
- Led interdisciplinary design reviews and technical evaluations.

RELEVANT PROJECTS

[EMC / EMI Analysis of MPV](#)

- Conducted EMC/EMI assessment for a multi-purpose vessel, reducing worst-case co-channel interference by over 18 dB.

[Antenna Deployer Module](#)

- Designed and validated CubeSat antenna deployment subsystem, passing rigorous thermal vacuum and communication tests.

[Collaborative PIC Design](#)

- Developed a high-performance silicon photonic integrated circuit that met stringent design specifications within 3 nm.

[WLAN for Huawei Drone](#)

- Proposed effective channel selection changes that reduced command link dropout events by 73% in urban areas.

EDUCATION

B.A.Sc. Electrical Engineering - [University of British Columbia, Vancouver](#) 2018 to 2023

Airline and Flight Operations Coursework - [BCIT](#) 2024, two terms

CERTIFICATIONS & LICENSES

- [Private Pilot License Training](#)** - Pacific Flying Club, Delta BC | 70+ hours on Cessna 152, flight-test competency focus
- FAA Part 107 UAS Pilot License**